# ABSTRACT

Agricultural products pass through several stages before reaching consumers, including cultivation, harvesting, processing, transportation, storage, and retail. Information generated at these stages is often maintained using separate systems, making it difficult to establish a reliable end-to-end product history and increasing the possibility of incorrect or manipulated records. Blockchain-based traceability improves record integrity, but does not guarantee the correctness of data before it is recorded.

This project proposes a blockchain-based agricultural product traceability system that introduces a data validation stage before records are committed to a permissioned blockchain network. The system maintains batch-level information throughout the supply chain and uses role-based access control for authorized stakeholders. Before an event is recorded, it is checked for completeness, authorization, duplication, and predefined business rules.

A QR code is generated for each agricultural batch, allowing consumers to retrieve product information by scanning the code. The system also captures selected sustainability indicators and combines their normalized values into a batch-level sustainability score.

The proposed framework integrates data validation, blockchain-based traceability, QR-based consumer verification, and sustainability assessment. The prototype will be evaluated using validation, blockchain performance, QR verification, and sustainability metrics.